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ORIGINAL ARTICLE

A comparative analysis of the impacts of climate change and irrigation on land surface and subsurface hydrology in the North China Plain

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Abstract Few studies have investigated the similar/different characteristics of regional environmental changes induced by climate change and human activities (e.g., irrigation, groundwater pumping). In this study, three climate change scenarios and one irrigation scenario were simulated using the Community Land Model 4.0 to investigate and compare the water-related environmental changes induced by climate change and irrigation in the North China Plain (NCP). Compared to future change climate scenarios, extensive groundwater-fed irrigation in the NCP could have similar magnitude of effects on land surface fluxes and states, but with much larger effects on subsurface water fluxes/states. For example, it was found that groundwater-fed irrigation has led to the decrease of annual mean water table depth by 1 m in major agricultural areas of the NCP while climate change has negligible impacts on water table depth. That is, human water use tends to dominate the subsurface water balance in the NCP. Moreover, climate change and irrigation exhibited different effects on the vertical profile of soil column. That is, irrigation appears to have much larger effects on the top layer soil moisture (SM) whereas increase in precipitation associated with climate change exerts more influence on lower layer SM. Through indentifying the similar/different characteristics between climate change and irrigation, our results highlight the importance of exactly accounting for the effects of human water use and could provide guidance for determining effective measures for adapting to environmental changes induced by climate change and human water use for this region.

Keywords Climate change · Irrigation · Land surface/subsurface hydrology · North China Plain

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Introduction

Water resources, among others, are important to many aspects of the environment, economy and society. It is well known that anthropogenic climate change have large impacts on moisture and heat transfer, with subsequent influence on precipitation and temperatures variations and change patterns (Trenberth et al. 2003). These changes in temperature and precipitation, in turn, could significantly alter land surface water balance through impacts on evaporation and surface runoff (Vörösmarty et al. 2000). Also, climate variability and change could affect subsurface hydrology both directly through aquifer recharge and indirectly through groundwater withdrawals (Taylor et al. 2012). Human water use, at the same time, were reported to have large impacts on the terrestrial water system, including aquifer storage through groundwater pumping and shallow soil moisture (SM) through irrigation. For example, around 70 % of global freshwater in 2000 was withdrawn for irrigation, which accounted for 90 % of consumptive water use (Döll 2002). This large-scale redistribution of freshwater from rivers, lakes, reservoirs and ground water to agricultural regions has significantly perturbed the local/regional water cycle (e.g., Haddeland et al. 2006; Tang et al. 2007; Kustu et al. 2010; Ozdogan et al. 2010; Lo and Famiglietti 2013). In semiarid and arid agricultural regions, groundwater withdrawals can even exceed groundwater recharge rate, resulting in persistent aquifer depletion when there are no alternative water sources (Scanlon et al. 2012; Rodell et al. 2009; Famiglietti et al. 2011; Leng et al. 2013b). In addition, irrigation was found to have large effects on the quality of water resources (e.g., increasing the salinity of receiving bodies), thus compromising the subsequent use of these waters for agricultural, domestic and industrial purposes (e.g., Causapé et al. 2004; Isidoro et al. 2006, 2010; Playán et al. 2008).

The North China Plain (NCP), a region home to more than 200 million people in China, is the most important center of agricultural production (Kendy et al. 2004). Because of the uneven distribution of rainfall, the agriculture development in the NCP depends heavily on irrigation (Wang et al. 2008a, b). Due to the shortage of surface water, especially in the northern part of the NCP, the main source of irrigation water has shifted from surface water to predominantly ground water since the 1980s (Liu et al. 2008). Groundwater over-exploitation, which resulted in land subsidence, desertification and large water table depression cones has since become a serious environmental issue for northern China (Wang et al. 2007). At the same time, the NCP was one of the most vulnerable regions to climate change due to increasing temperature and water-related eco-environmental issues (Xiong et al. 2007; Mo et al. 2009). A rainfall-decreasing trend, which has been observed for the last two decades, has greatly affected the water resources availability and the groundwater recharge in the area. The NCP has become one of the most vulnerable regions to climate change and human perturbations in China due to its increasing climate variability (Fu et al. 2009; Leng et al. 2014) and strong dependence on irrigation for crop production with limited water resources (Mo et al. 2013).

Using the Variable Infiltration Capacity model, Dan et al. (2012) investigated the sensitivities of land surface hydrology to various hypothetic climate change scenarios in the Huang-Huai-Hai plain (covering most of the NCP), but the impacts on ground water, which is the primary water source for agriculture, industry and domestic use of this region, have not been examined. By implementing a simple irrigation scheme in Community Land Model version 3.5, Zou et al. (2014) investigated the effects of groundwater exploitation on land surface and subsurface fluxes/states in Hai River basin of China. In their irrigation scheme, however, the irrigation amount calculations were based on empirical functions, so interannual and seasonal variability of irrigation is not represented, which could lead to overestimation or underestimation of irrigation effects. Moreover, information on the sources of irrigation water was also not effectively accounted for in their model scheme. These two aspects of deficiencies which are common in most previous irrigation modeling studies (e.g., Haddeland et al. 2006; Sacks et al. 2009; Lobell et al. 2009; Ozdogan et al. 2010; Kueppers et al. 2007) could lead to misrepresentation of irrigation effects and thus inhibit the exact investigation of irrigation impacts at the local and regional scale (Sorooshian et al. 2012; Leng et al. 2013a, b). In addition to separately and exactly assessing the effects of climate change and human perturbations on the terrestrial water system, understanding the similar/different roles that climate change and human water use play in the regional water cycle can greatly facilitate the development of sustainable water management strategies (Ferguson and Maxwell 2012). This is especially true for the NCP, which is one of the most vulnerable regions to climate change and human perturbations in China.

Motivated by the above needs, the primary objective of this study is to investigate and indentify the similar/different effects of climate change and groundwater-fed irrigation on the land surface and subsurface hydrology in the NCP, which covers 34° N–41° N, 113° E–121° E, as shown in Fig. 1a.

Data and methodology

Study area

Figure 1a show the modeling domain in this study. This case profile includes Beijing and Tianjin municipalities together with most part of Hebei, northern Shandong and northern Henan provinces. This semiarid area is characterized by cold dry winters and hot humid summers, which is strongly affected by the East Asian Monsoon (Liu et al. 2001). The soil is characterized by the fertile topsoil and plenty of organic matter in the loam soil (Sun et al. 2006). The main land uses in this region are winter-wheat and summer-maize with mean annual evapotranspiration up to 850 mm, which is far in excess of the long-term average annual precipitation of about 550 mm (Foster et al. 2004; Moiwo et al. 2010). Therefore, irrigation is the key for ensuring food production in this region (Mo et al. 2013). With population increases and economic development, the excessive groundwater pumping has led to a rapid and consistent depletion of groundwater resources in many places across the region (Liu et al. 2001; Wang et al. 2007).

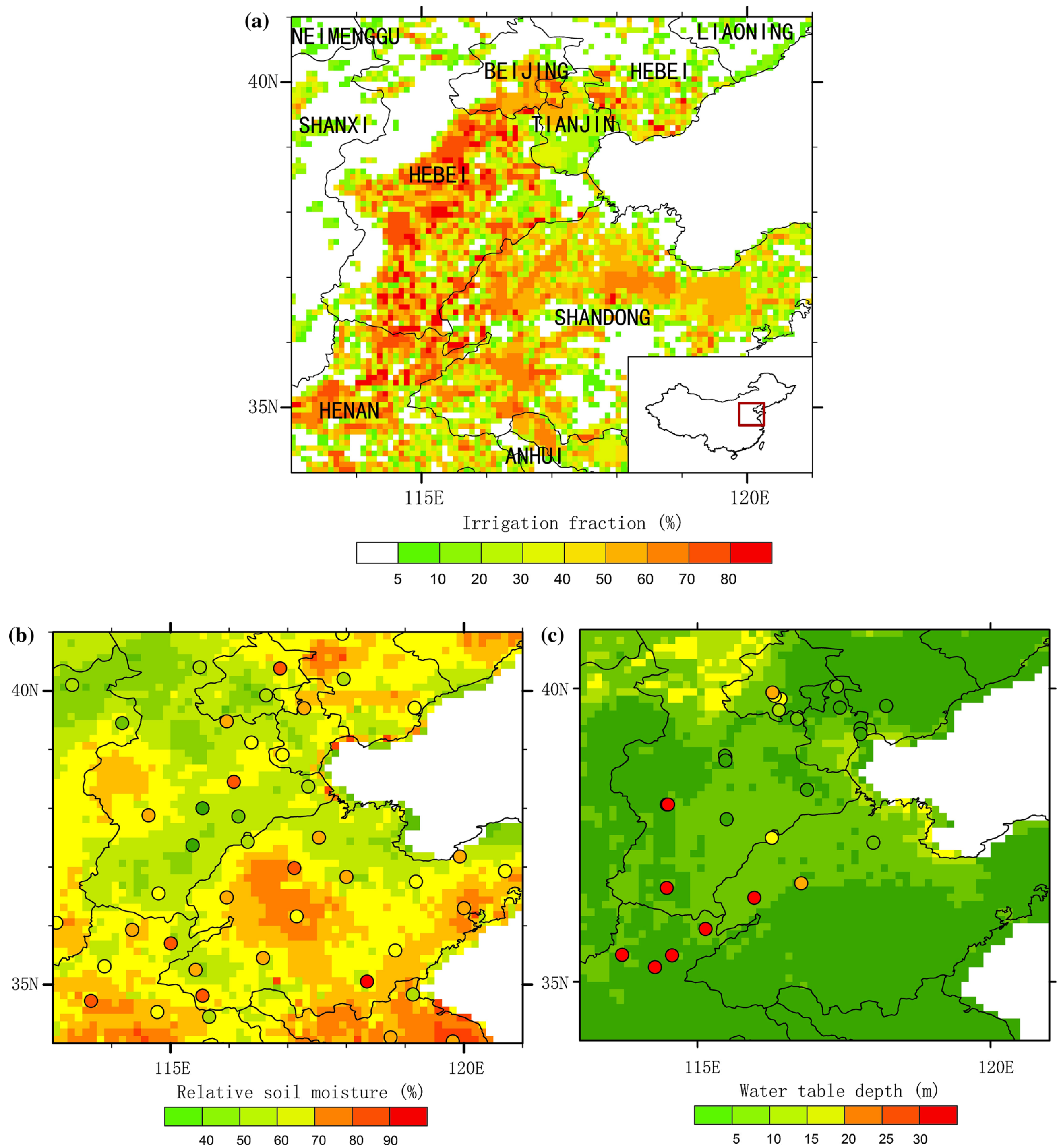


Fig. 1 **a** Model domain of this study. The *colors* represent the percentage of irrigated area in a grid cell from Siebert et al. (2005). **b** Comparison of summer mean relative SM (%) for top 0–10 cm layer for the period 1993–2002 and **c** annual mean water table depth (m) for the period 2004–2006 by the CTRL and observations (color figure online)

Data

High-quality atmospheric forcing datasets are of great significance for land surface modeling studies. Here, we use the assimilated forcing dataset from Chen et al. (2011) (http://westdc.westgis.ac.cn/data/7a35329c-c53f-4267-aa07-e0037d913a21). This datasets contains air temperature, precipitation rate, downward shortwave and longwave radiations, specific humidity, air pressure and wind speed with temporal resolution of 3-h from 1980 to 2010, at a

Table 1 Actual versus estimated irrigation amounts in 2000 and estimated model parameters

Administrative regions	Actual irrigation amounts (10^8 m^3)	Simulated irrigation amounts (10^8 m^3)	Weighted factor (f_{irrig})	Percent of groundwater withdraws (r_{grd}) (%)	Percent of surface water withdraws (r_{srf}) (%)
Beijing	17.16	17.43	0.5	70	30
Tianjin	9.86	14.88	0.5	45	55
Hebei	153.60	137.05	0.9	81	19
Shandong	167.32	181.29	0.8	53	47
Henan	150.12	139.53	0.8	58	42

$0.1° \times 0.1°$ grid resolutions across the continental China. This dataset was produced by merging a variety of data sources and was well suited for driving land surface models and other terrestrial modeling systems (Yang et al. 2010; Chen et al. 2011).

The irrigation fraction map was obtained from Siebert et al. (2005). This map representing the fraction of irrigated area at a resolution of 5 arc minutes around the year of 2000 has become the de facto information source for spatial distribution of global irrigated areas. In general, irrigation fractions are high in the piedmont plains around Taihang Mountain and low in the coastal areas of the NCP (Fig. 1a).

Water use data at the provincial level from the Beijing Water Resource Bulletin of 2000, Tianjin Water Resource Bulletin of 2000, Hebei Water Resource Bulletin of 2000, Henan Water Resource Bulletin of 2000 and Shandong Water Resource Bulletin of 2000 were collected. The total water withdrawals for irrigation were used for model calibration. The groundwater and surface water withdrawals and total water withdrawals were used to estimate the ratios of irrigation withdrawals from surface water (hereafter denoted as r_{surf}) and groundwater (hereafter denoted as r_{grd}), respectively, as follows:

$$r_{surf} = \frac{irr_{surf}}{irr_{total}} \tag{1}$$

$$r_{grd} = \frac{irr_{grd}}{irr_{total}} \tag{2}$$

where irr_{surf}, irr_{grd} and irr_{total} are the amount of surface water, ground water and total water withdrawals for irrigation, respectively. The r_{suf} and r_{grd} were estimated for each province, the native resolution of the data sources. The grid cells in each province were assigned the estimated r_{suf} and r_{grd} and remain constant in our modeling period. Surface water is the primary source for irrigation in Tianjing with r_{srf} up to 55 %, while groundwater serves as the primary source for irrigation in the Beijing, Hebei, Henan and Shandong provinces with r_{grd} up to 70, 81, 53 and 58 % (see Table 1), respectively.

The relative SM for the shallow soil layer (0–10 cm) was obtained from the National Meteorological Information Center of the China Meteorological Administration (http://cdc.cma.gov.cn/home.do). The SM was measured using the gravimetric technique and recorded as mass percentage with a 10-day interval (i.e., on the 8th, 18th and 28th of every month). The summer (June, July and August) mean monthly values are converted to relative SM and used to evaluate the model performance since the data were recorded mostly in the warm season with no measurements made over frozen soil. For model validation, we consider the period 1993–2002 with the most available records in the NCP due to sparse observations available in other years (Li et al. 2005). Sites with at least one 10-day interval of missing data during the summer period were excluded. As a result, a total of 47 stations with complete records were selected for use.

A collaborative project was initiated recently among the Institute of Geographic Sciences and Natural Resources Research of Chinese Academy of Sciences, National Institute for Environmental Studies of Japan and the Tianjin Institute of Geology and Mineral Resources of China to monitor the groundwater levels in the Yellow River Basin including the NCP (Wang et al. 2009). Within the framework of this project, automatic monitoring equipments (KADEC-MIZU II) were used to monitor the dynamics of groundwater at a frequency of 30 min or 1 h. In this study, data from a total of 28 well maintained observation wells with longer records during the period 2004–2006 in the NCP were obtained. For ground water validation, we compared the annual mean water table depth simulations with the observed annual mean values for the period 2004–2006. We considered the annual mean values for our use because the variance and the variation coefficient of observations are large (Wang et al. 2009).

Model description and setup

The Community Land Model version 4.0 (CLM4) was used in this study. The CLM4 is a land surface model and can be used in coupled and offline modes in the framework of the Community Earth System Model (Gent et al. 2010; Lawrence et al. 2011). In CLM4, extensive enhancements were represented in hydrological processes such as runoff generation, soil hydrology and groundwater dynamics (Lawrence et al. 2011). CLM4 includes an irrigation scheme that computes energy and plant processes for irrigated cropland (Leng et al. 2013a, b). The cropland area of each grid cell is divided into an irrigated and unirrigated fraction based on a dataset of areas equipped for irrigation (Siebert et al. 2005). In irrigated croplands, a check is made at 6:00 a.m. each day to determine whether irrigation is required on that day. Irrigation is triggered when the crop leaf area >0, and $\beta_t < 1$ (i.e., water is limiting photosynthesis). The target SM content for each soil layer ($W_{\mathrm{target},i}$, kg m^{-2}) is expressed as follows:

$$W_{\mathrm{target},i} = (1 - f_{\mathrm{irrig}}) \times W_{o,i} + f_{\mathrm{irrig}} \times W_{\mathrm{sat},i} \quad (3)$$

where $W_{o,i}$, is the minimum SM content, preventing water stress for crop growth; $W_{\mathrm{sat},i}$ is the SM content at saturation in that layer; f_{irrig} is a weighted factor between 0 and 1, corresponding to setting the SM target just enough to result in no water stress for crops or to full soil saturation. The irrigation amounts defined by the deficit between the current SM content and the target SM content was applied evenly to the soil beneath the irrigated crops during 6:00–10:00 a.m. In the irrigation scheme of CLM4, the simulated irrigation amount is removed from the total liquid runoff (i.e., q_{runoff}) to simulate the removal from local rivers within each grid cell, which is not realistic as most previous studies especially for regions rely heavily on groundwater source such as the NCP. Therefore, we implemented an operational groundwater pumping scheme for CLM4 following Leng et al. (2013b) in this study to mimic the irrigation process more realistically.

Experimental design

In this study, five offline CLM4 simulations were performed and summarized in Table S1. Briefly, the control simulation (i.e., CTRL) was conducted to simulate the natural state, ignoring climate change and human impacts. In IRRIG, the simulated irrigation amount was set to be withdrawn from the ground water and surface water proportionally following the groundwater pumping scheme as discussed in section "Model description and setup." In addition to accounting for irrigation water sources, accurate estimation of water amounts applied for irrigation is crucial for irrigation modeling studies (Sorooshian et al. 2012; Leng et al. 2013a). In this regard, we calibrated the model by perturbing f_{irrig} between (0, 1) at a regular interval of 0.1 following Leng et al. (2013a) to match the observations at the provincial level. It should be noted that the model calibration for Henan and Shandong provinces were done for the entire administrative district although our study domain cover a portion of Henan and Shandong provinces (Fig. 1) due to the constraint of data availability at finer scale. The difference between IRRIG and CTRL corresponds to the effects of irrigation.

Three hypothetical climate scenarios were designed to account for possible future climate changes relative to the baseline of 1980–2000 in the NCP. These climate scenarios were based on future climate projections by 2050 which are likely to be experienced from more than 40 global models (Qin et al. 2005). Here, the climate change scenarios were implemented by perturbing the baseline meteorological forcing to reflect the median projected temperature change and a range of projected precipitation change by year 2050. More specifically, the scenarios were derived by (1) systematically increasing air temperature by 2 °C while all other meteorological forcings were unchanged (i.e., HOT), (2) same with HOT but with a 15 % decrease in precipitation (i.e., HOTDRY), and (3) same with HOT but with a 15 % increase in precipitation (i.e., HOTWET). The differences between the climate change scenarios (i.e., HOT, HOTDRY and HOTWET) and CTRL were used to evaluate the potential effects of future climate change on regional hydrology in the NCP. We focus our analysis on the variations in the amplitude of projected hydrological changes, due to the limitations of the change factor approach which did not change the characteristics and the spatial variability of climate variables.

Land surface parameters such as PFTs (Plant Functional Types), LAI (Leaf Area Index), SAI (Stem Area Index) and land cover composition of the grid cells were derived from the 0.05° CLM4 input dataset developed by Ke et al. (2012). All gridded datasets mentioned in section "Data" were mapped to 0.125° resolution as inputs to CLM4. Before the simulations, an offline spin-up was performed by cycling the meteorological forcing for 20 cycles using the CRTL setup until equilibrium conditions were achieved for all state variables including SM, temperature and groundwater table depth. The resulting state variables were used as the initial condition for all simulations.

Results

Model validation and analysis

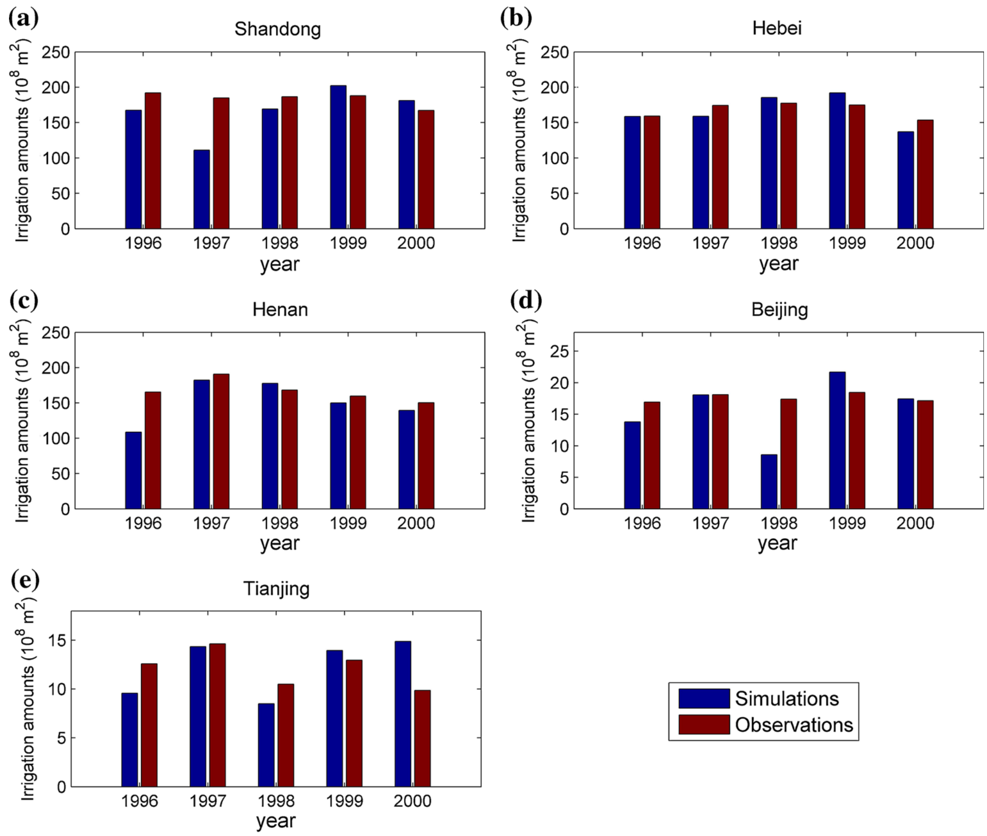
Figure 1b, c shows the spatial distributions of relative SM and groundwater table depth by the CTRL and observations. It should be noted that the summer mean SM for the period 1993–2002 and the annual mean simulated water table depth for the period 2004–2006 are used in Fig. 1b, c respectively. The specific periods selected for model validations were mainly constrained by the available records we can obtain in the study area. Also, our modeling results can be validated only in a general manner due to the scale mismatch between simulations and observations. In general, the control simulation by CLM4 performs well in capturing the spatial pattern of relative SM (Fig. 1b), which ranges from 32 to 95 % over the whole region and decreases from south to north consistent with the observations. The highest SM in CTRL was located near the Taishan Mountain area (centered at 117 E and 36.5 N) in the Shandong province. Relatively lower SM was found over agricultural regions when compared to observation data. The relative bias of simulated relative SM against observations was about −13 % (note: the mean value of simulations was calculated by averaging the values of grid cells closest to the observation sites). The water table depths correlate well with the regional climatology. Specifically, the northwestern grassland and desert regions have deeper groundwater, while the water table is shallower (close to the land surface) in the southeast humid regions. Due to the absence of human activities in CTRL, much shallower water table depth was simulated compared to observations in the southern NCP, where extensive groundwater-fed irrigation exists (Fig. 1a). As shown in Fig. 1c, the simulated water table depth was below 10 m over major parts of the NCP, while the observed groundwater is far deeper and can even exceed 30 m in some areas in northern Henan and southern Hebei provinces which rely heavily on groundwater source for irrigation. The relative bias of simulated water table depth against observations was −40 % (note: the mean value of simulations was calculated by averaging the values of grid cells closest to the observation sites). The significant underestimation of water table depth indicates the importance of accounting for human water use in this region.

In Table 1, the reported and simulated irrigation amounts after calibration in 2000 for Beijing, Tianjin, Hebei, Henan and Shandong provinces were presented. Although some discrepancy exists between the actual and simulated irrigation amounts, this represents the best model performance with calibration given the available datasets at the coarse resolution (i.e., provincial scale). Irrigation modeling could be better constrained at higher resolutions (Leng et al. 2013b). The ability of CLM4 after calibration to capture the interannual variability of irrigation amounts was also demonstrated in Fig. 2, by comparing the annual irrigation amounts simulated by the IRRIG with reported annual irrigation amounts for the consecutive period 1996−2000 for each administrative unit in the study area. Using the calibrated and estimated parameters (i.e., f_{irrig}, r_{surf} and r_{grd}) in Table 1, the spatial distribution of irrigation amounts by the IRRIG was shown in Fig. 3a. High and low irrigation amounts were concentrated around Taihang Mountain and in the coastal regions, respectively, consistent with the spatial pattern of the irrigated fractions. Irrigation amounts up to 30 mm month^{-1} were found over most of southern Hebei province and northern Henan province due to the relatively arid climate (Fig. 3a). When the simulated irrigation amounts by the IRRIG was met by withdrawing from groundwater based on the ratio of r_{grd}, the water table depth increased significantly compared to the simulations by the CTRL (Fig. 3b). Importantly, the simulated water table depth by the IRRIG matched the observations more closely, with the relative bias of −14 %. The underestimation of water table depth against observations is reasonable given the fact that groundwater pumping for domestic and industrial water use are not included in the model. Overall, the CLM4 with the groundwater pumping scheme after calibration is well suited for modeling irrigation effects in our following analyses.

Spatial distribution of the effects by irrigation and climate change on surface and subsurface hydrology

Groundwater-fed irrigation can perturb the surface water and energy balances induced by climate change through changes in groundwater recharge, SM, ET and other budget terms. Figure 4 shows the spatial distributions of the difference of annual mean water table depth (Z_{∇}, m), top layer SM (mm^3 mm^{-3}) and latent heat flux (LH, W m^{-2}) between each scenario and the CTRL. The Summary of the difference of annual mean Z_{∇}, SM and LH between each scenario and the CTRL for the NCP was also shown in Table S2. When accounting for the source of irrigation withdrawals, groundwater-fed irrigation reduces water table levels greatly over the agricultural regions due to long-term exploitation as shown in Fig. 4a. That is, increase of recharge induced by additional irrigation water was much lower compared to the amounts of groundwater withdrawals, leading to the deeper groundwater table with an annual drop of Z_{∇} by 0.4 m averaged over the region. However, the regions over northern Shandong province do not suffer significant drop of groundwater table, partly due to the low irrigation amounts (see Fig. 3a). At locations relying heavily on the groundwater source (e.g., southern Hebei province), the annual mean Z_{∇} was simulated to drop by more than 1.5 m at the time of maximum extraction. Compared to the impacts of irrigation, the effects of climate change on Z_{∇} show a relatively uniform distribution throughout the NCP, with values mostly between −0.06 to 0.12 m. In HOT, more water evaporated before infiltrating and recharging the aquifer storage, resulting in widespread decrease of annual mean Z_{∇} by 0.02 m over a major portion of the study region relative to the baseline condition (i.e., CTRL). Compared to HOT, much larger decrease of groundwater levels by 0.1 m was found in HOTDRY. This could be attributed to the decrease of

Fig. 2 Comparison of the annual irrigation amounts simulated by the IRRIG with reported annual irrigation amounts for the consecutive period 1996–2000 for each administrative unit in the study area



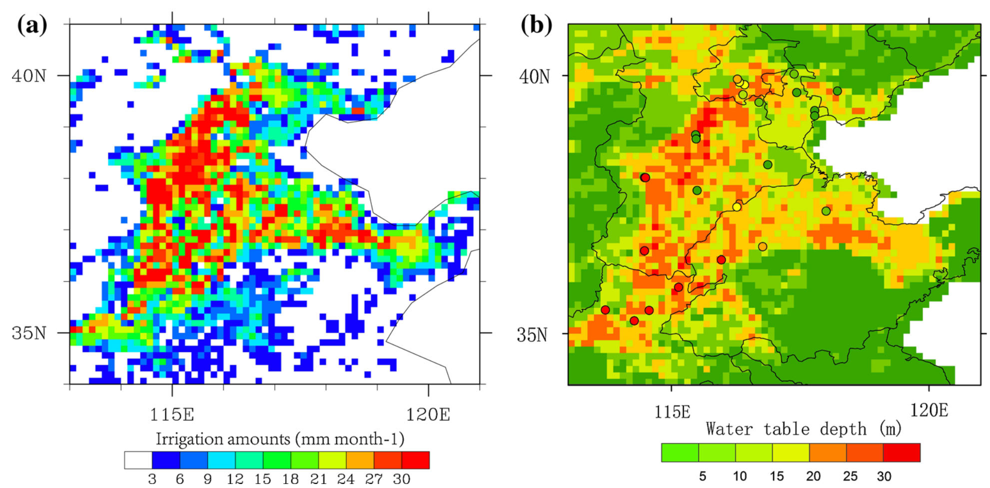


Fig. 3 **a** Spatial distribution of irrigation amounts (mm/month) simulated by the IRRIG and **b** comparison of modeled annual mean water table depth (m) by the IRRIG with observations for the period 2004–2006

precipitation over the water-limited semiarid region which is most sensitive to precipitation changes. Conversely, annual mean Z_∇ increased by 0.04 m due to increased recharge to the aquifer system over most of the NCP from HOTWET. Overall, the changes of annual mean Z_∇ induced by groundwater-fed irrigation were much larger than those by climate changes across the NCP.

Because the irrigation water withdrawn from surface water and groundwater was applied to the soil, significant increase of SM was detected. From IRRIG-CTRL, irrigation increases the SM by 0.04 mm^3 mm^{-3} over agricultural regions, with some grid cells in Hebei province experiencing a more evident increase by 0.06 mm^3 mm^{-3}. Widespread decrease of SM was found over the study

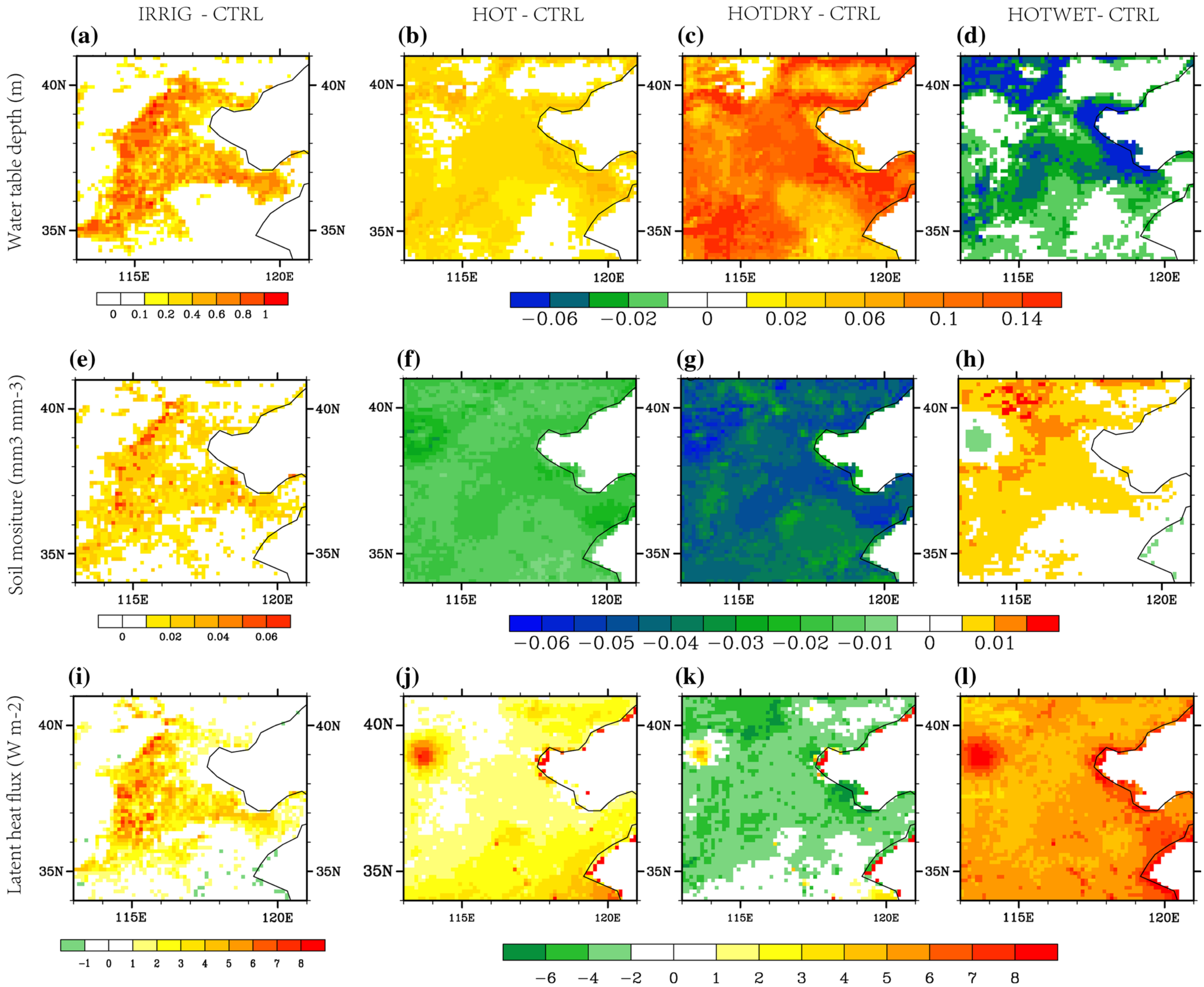


Fig. 4 Spatial distributions of annual mean differences for water table depth (Z_{∇}, m), top layer SM ($mm^3\ mm^{-3}$) and latent heat flux (LE, ($W\ m^{-2}$) between each scenario and the CTRL

domain due to enhanced evapotranspiration (ET) in the HOT scenario. From HOTDRY-CTRL, much drier SM was found due to reduced precipitation and enhanced ET, with SM decreasing by more than 0.05 $mm^3\ mm^{-3}$ across the NCP. With increased precipitation and temperature in HOTWET, however, SM changes depend on the combined effects of the water and heat budgets. From HOTWET-CTRL, SM shows a slight increase of 0.01 $mm^3\ mm^{-3}$ over major areas of the study domain due to the increased precipitation. Conversely, an obvious decrease of SM was found around the area (38 N–40 N, 113 W–115 W) where enhanced ET in a warmer climate exceeds the water input by the increased precipitation. Generally, the areas of increased SM are more extensive than the areas of decreased SM. This implies that the enhanced water input through increased precipitation tends to dominate over the water loss through increased ET and leads to a general wetting of soils over the NCP in the HOTWET scenario.

The wetting effects induced by irrigation were also reflected by the changes in LH. Irrigation increases moisture availability over agricultural areas, increasing LH from land evaporation and plant transpiration. From IRRIG-CTRL (Fig. 4e), significant increase of LH was found in southern Hebei, northern Henan and western Shandong provinces where large-scale irrigation practices exist. Specifically, annual mean LH increased by 4 $W\ m^{-2}$ over major areas of agricultural regions in the NCP, which are consistent with previous studies (e.g., Lobell et al. 2009; Ozdogan et al. 2010). In some areas in southern Hebei province, LH increased by more than 8 $W\ m^{-2}$ in the summer of dry years. In general, similar magnitude of differences was found between the climate change and irrigation scenarios.

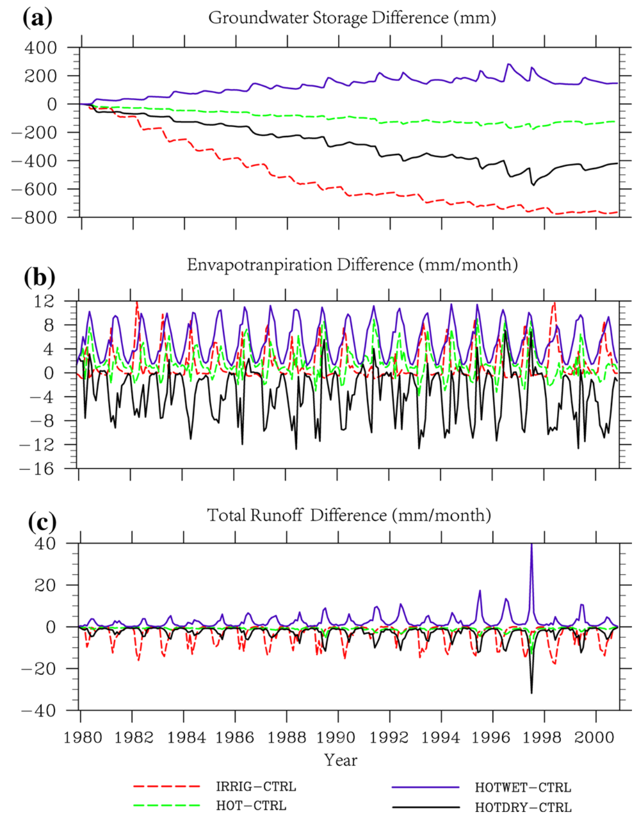


Fig. 5 Temporal variations of the differences for groundwater storage **a** evaportranspiration **b** and total runoff **c** between each scenario and the CTRL. Note the values are area-weighted averages over the NCP defined in Fig. 1a

Temporal variations and vertical profile of the effects by irrigation and climate change on surface and subsurface hydrology

Figure 5 shows the temporal variations of the difference of groundwater storage [W_t (mm)], ET (mm) and total runoff between each scenario and the control simulation averaged spatially over the study domain in Fig. 1a. In IRRIG-CTRL, groundwater pumping reduces aquifer storage persistently throughout the time horizon, with larger decreases in dry years when more groundwater was withdrawn to meet crop demand. As shown in Fig. 5a, groundwater storage has decreased by 800 mm during 1980–2000 due to groundwater over-exploitation. In addition, the effects by IRRIG on W_t exhibit distinct seasonal variability with greater effects in the growing season. Compared to the effects on W_t by IRRIG, much smaller impacts on W_t were found in the HOTDRY and HOT scenarios relative to the 1980–2000 baselines. Specifically, W_t decreased by 400 and 100 mm in HOTDRY-CTRL and HOT-CTRL, respectively. With precipitation increased by 15 % in HOTWET, W_t increased by more than 200 mm relative to the 1980–2000 baselines due to increased SM and infiltration, which further facilitated recharge to the aquifer storage.

As expected, irrigation increased the SM for crop demand, which resulted in persistent increase of ET by 4 mm month^{-1} in the growing season. From Fig. 5b, the effects on ET by IRRIG and HOT exhibited similar magnitude. However, in relatively dry years (e.g., 1998) when more irrigation water was required to meet the crop demands, much larger increase of ET was simulated in IRRIG compared to HOT. With energy and precipitation increase combined, ET increased by more than 10 mm month^{-1} in the NCP as shown in HOTWET-CTRL.

Wetter SM may lead to more surface runoff to be generated, while deeper groundwater table could constrain subsurface runoff generation. The total runoff changes depend on the combined changes of surface runoff and subsurface runoff. As shown in Fig. 5c, groundwater-fed irrigation has led to persistent decline of total runoff averaged over the NCP, suggesting that limiting subsurface generation by groundwater pumping could outweigh the enhancement of surface runoff generation by irrigation return flow. This may be due to the fact that subsurface runoff is parameterized as an exponential function of water table depth in CLM4 (Niu et al. 2005). From HOT-CTRL, decrease of total runoff was found, but with a negligible magnitude except in dry years (e.g., 1997). With decrease of precipitation and increase of temperature combined in HOTDRY, the soil becomes much drier, which lowered surface runoff generation and subsequently resulting in decrease of the total runoff by 5 mm month^{-1}. Conversely, the soil became wetter due to additional water input in HOTWET, which has led to the enhancement of surface runoff generation. Generally, changes in precipitation had large impacts on the changes in annual and seasonal runoff, while changes in temperature had minor effect on annual runoff in the NCP. Moreover, groundwater exploitation has exerted much greater control on runoff than precipitation in the NCP.

It is interesting to note from the comparison of Fig. 4e, h that the wetting effects induced by irrigation (i.e., IRRIG) show similar spatial structures to that by HOTWET. To further look into whether the pattern of changes in SM content have similar vertical structure in the two scenario simulations, the vertical profiles of moisture content changes in the ten soil layers were shown in Fig. 6. As expected, the whole soil column becomes wetter and show pronounced seasonal and interannual variations due to irrigation (Fig. 6a). The changes of SM, however, were mainly found in the top soil layers (i.e., from soil layer one to five), with minor increase on the lower soil layers. This shows that despite more water was added by irrigation, SM in the lower layers was not significantly enhanced due to

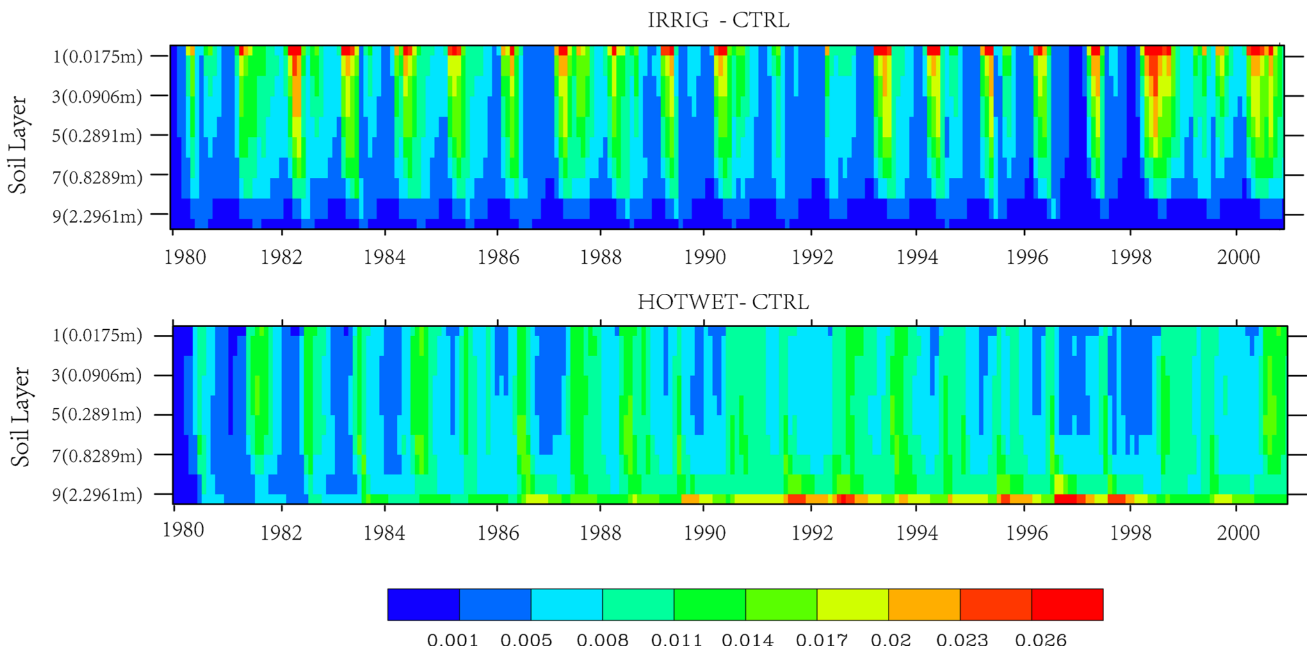


Fig. 6 Vertical profile of moisture content changes ($mm^3\ mm^{-3}$) for the ten soil layers by IRRIG-CTRL (*top panel*) and HOTWET-CTRL (*bottom panel*). The numbers in the *bracket* show the soil depth (m) from ground to the specific soil layer in the CLM4

recharge to the aquifer system to balance the removal of large amounts of water by groundwater pumping. In HOTWET, SM increased by 0.011–0.17 $mm^3\ mm^{-3}$, which was distributed more uniformly throughout the whole soil column, with pronounced increase of 0.023 $mm^3\ mm^{-3}$ found in relatively wet year (e.g., 1992). Since precipitation was increased by 15 % uniformly over time, this allowed water to infiltrate through the soil layers and enhance the groundwater storage. This resulted in larger increases in SM content in lower layers and shallower water table depth. In summary, governed by water table depth through groundwater–land surface interactions, the effects of irrigation appear to be more concentrated on top soil layers whereas the effects of climate change, prescribed by a uniform increase in precipitation over time, was much larger on the lower soil layers.

Summary and conclusions

In this study, the effects of climate change and irrigation on surface/subsurface hydrology were investigated and compared in the NCP, which is one of the most vulnerable regions to climate change and human perturbations in China. Major findings of this research were summarized as below:

Our results show that groundwater exploitation, which removed a large amount of groundwater from aquifer and applied to the soil column, has significantly altered the water availability and distribution in the NCP. Without explicitly accounting for irrigation using an interactive scheme (i.e., irrigation amounts and irrigation source were not determined dynamically by SM conditions), Ferguson and Maxwell (2012) showed that groundwater pumping and irrigation could have similar impacts on surface water and groundwater resources to a warming climate. However, our results show that compared to change in climate scenarios, extensive groundwater-fed irrigation in the NCP could have similar magnitude of effects on land surface fluxes and states (e.g., ET, LH, and shallow-layer SM), but with much larger effects on subsurface water fluxes/states (e.g., Z_{∇} and W_t). For example, it was found that groundwater-fed irrigation has led to the decrease of annual mean water table depth by 1 m in major agricultural areas of the NCP while climate change has much smaller impacts on water table depth. Our results also emphasize the need to accurately account for irrigation water source and water amounts in irrigation modeling studies.

The effects of irrigation on W_t, ET and total runoff exhibit distinct seasonal variability with greater effects in the growing season, while the effects of climate change were more uniform over time. Moreover, governed by local water table depth through groundwater–land surface interactions, climate change and irrigation could play different role in modulating changes in the vertical profile of SM contents. That is, the impacts of agricultural irrigation on SM are mostly confined to the top soil layers while precipitation changes that may occur uniformly over time may have more influence on the lower soil layer moisture. These differences have great implications for developing adaptation and mitigation strategies for sustainable water resources.

However, our climate change results are based on hypothetical scenarios with spatially uniform and temporally constant increase in temperature and percentage change in precipitation. Changes in precipitation characteristics such as frequency and intensity may lead to changes in SM profiles different from the present results. Also, our irrigation results are based on spatially uniform and temporally constant parameters (i.e., f_{irrig}, r_{grd} and r_{srf}) which were calibrated and estimated at the provincial level due to data (i.e., census) limitations at a finer scale. Moreover, the modeled results could be sensitive to the choice of irrigation map as investigated by Leng et al. (2013a). The uncertainty arising from irrigation map could be addressed either by accurate representation of the spatial distribution and intensity of irrigated areas at the first place or by model calibration as adopted in this study (Leng et al. 2013a). More efforts should be made to develop a high spatial–temporal dataset to eliminate the limitations and the uncertainty arising from these dataset.

It should also be noted that the impacts by human activities and climate could, to some extent, be interrelated. For example, anthropogenic climate change could affect human water use through changes in water availability and water demand (Döll 2002). In turn, human water use could also exert significant influence on local/regional climate through increased ET (Kueppers et al. 2007; Lobell et al. 2009; Sacks et al. 2009). The interactions between climate change and irrigation could be particularly significant if climate change leads to reduced precipitation in agricultural regions that already rely heavily on groundwater-fed irrigation. Reduced precipitation would increase irrigation demand and further lower the groundwater table by irrigation, as simulated in this study. Through surface water–groundwater interactions, this could potentially reduce ET and enhance the drying trend. An assessment of the local hydrological response to these two-way dynamical interactions is lacking and beyond the scope of this study. Also, the joint impacts of climate changes and irrigation on local water resources especially on groundwater can be addressed through driving the calibrated CLM4 with the physically constrained climate projections by general climate models in the future.

Despite the uncertainties and deficiency of this study, our approach of assessing the separate impacts of climate change and irrigation on surface/subsurface hydrology in a spatially consistent manner is the first of its kind in the NCP. Through identifying the similar/different characteristics between climate change and irrigation, our results emphasize the need to exactly investigate the impact of human actives since human water use tends to dominate the subsurface water balance in NCP and could provide guidance for effectively determining adaptation and mitigation strategies to environmental changes induced by future climate change and human water use.

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References

Causapé J, Quílez D, Aragüés R (2004) Assessment of irrigation and environmental quality at the hydrological basin level: II. Salt and nitrate loads in irrigation return flows. Agr Water Manage 70(3):211–228. doi:10.1016/j.agwat.2004.06.006

Chen Y, Yang K, He J, Qin J, Shi J, Du J, He Q (2011) Improving land surface temperature modeling for dry land of China. J Geophys Res 116:D20104. doi:10.1029/2011JD015921

Dan L, Ji J, Xie Z, Chen F, Wen G, Richey JE (2012) Hydrological projections of climate change scenarios over the 3H region of China: a VIC model assessment. J Geophys Res 117:D11102. doi:10.1029/2011JD017131

Döll P (2002) Impact of climate change and variability on irrigation requirements: a global perspective. Clim Chang 54(3):269–293. doi:10.1023/A:1016124032231

Famiglietti JS, Lo M, Ho SL, Bethune J, Anderson KJ, Syed TH, Swenson SC, de Linage CR, Rodell M (2011) Satellites measure recent rates of groundwater depletion in California's Central Valley. Geophys Res Lett 38:L03403. doi:10.1029/2010GL046442

Ferguson I, Maxwell RM (2012) Human impacts on terrestrial hydrology: climate change versus pumping and irrigation. Environ Res Lett 7:044022. doi:10.1088/1748-9326/7/4/044022

Foster S, Garduño H, Evans R, Olson D, Tian Y, Zhang W, Han Z (2004) Quaternary aquifer of the North China Plain—assessing and achieving groundwater resource sustainability. Hydrogeol J 12:81–93

Fu G, Charles SP, Yu J, Liu C (2009) Decadal climatic variability, trends, and future scenarios for the North China Plain. J Clim 22:2111–2123. doi:10.1175/2008JCLI2605.1

Gent P, Yeager S, Neale R, Levis S, Bailey D (2010) Improvements in a half degree atmosphere/land version of the CCSM. Clim Dyn 34:819–833. doi:10.1007/s00382-009-0614-8

Haddeland I, Lettenmaier DP, Skaugen T (2006) Effects of irrigation on the water and energy balances of the Colorado and Mekong river basins. J Hydrol 324:210–223. doi:10.1016/j.jhydrol.2005.09.028

Isidoro D, Quílez D, Aragüés R (2006) Environmental impact of irrigation in La Violada district (Spain). J Environ Qual 35(3):766–775

Isidoro D, Quílez D, Aragüés R (2010) Drainage water quality and end-member identification in La Violada irrigation district (Spain). J Hydrol 382:154–162. doi:10.1016/j.jhydrol.2009.12.026

Ke Y, Leung RL, Huang M, Coleman AM, Li H, Wigmosta MS (2012) Developing high resolution land surface parameters for community land model. Geosci Model Dev 5:1435–1481. doi:10.5194/gmdd-5-1435-2012

Kendy E, Zhang Y, Liu C, Wang J, Steenhuis TS (2004) Groundwater recharge from irrigated cropland in the North China Plain: case study of Luancheng County, Hebei Province, 1949–2000. Hydrol Process 18:2289–2302

Kueppers LM, Snyder MA, Sloan LC (2007) Irrigation cooling effect: regional climate forcing by land-use change. Geophys Res Lett 34:L03703. doi:10.1029/2006GL028679

Kustu MD, Fan Y, Robock A (2010) Large-scale water cycle perturbation due to irrigation pumping in the US High Plains: a synthesis of observed streamflow changes. J Hydrol 390(3–4):222–244. doi:10.1016/j.jhydrol.2010.06.045

Lawrence D, Oleson KW, Flanner MG, Thorton PE, Swenson SC, Lawrence PJ, Zeng X, Yang Z, Levis S, Skaguchi K, Bonan GB, Slater AG (2011) Parameterization improvements and functional and structural advances in version 4 of the community land model. J Adv Model Earth Syst 3:M03001. doi:10.1029/2011MS000045

Leng G, Huang M, Tang Q, Sacks WJ, Lei H, Leung LR (2013a) Modeling the effects of irrigation on land surface fluxes and states over the conterminous United States: sensitivity to input data and model parameters. J Geophys Res 118:9789–9803. doi:10.1002/jgrd.50792

Leng G, Huang M, Tang Q, Leung LR (2013b) Modeling the effects of groundwater-fed irrigation on terrestrial hydrology over the conterminous United States. J Hydrometeorol. doi: 10.1175/JHM-D-13-049.1

Leng G, Tang Q, Huang M, Hong Y, Leung LR (2014) Projected changes in mean and interannual variability of surface water over continental China. Sci China Earth Sci (Accepted)

Li H, Robock A, Liu S, Mo X, Viterbo P (2005) Evaluation of reanalysis soil moisture simulations using updated Chinese soil moisture observations. J Hydrometeorol 6:180–193

Liu C, Yu J, Kendy E (2001) Groundwater exploitation and its impact on the environment in the North China Plain. Water Int 26:265–272

Liu J, Zheng C, Zheng L, Lei Y (2008) Ground water sustainability: methodology and application to the North China Plain. Ground Water 46:897–909

Lo M, Famiglietti JS (2013) Irrigation in California's Central Valley strengthens the southwestern US. Water cycle. Geophys Res Lett 40:301–306. doi:10.1002/grl.50108

Lobell D, Bala G, Mirin A, Phillips T, Maxwell R, Rotman D (2009) Regional differences in the influence of irrigation on climate. J Clim 22:2248–2255. doi:10.1175/2008JCLI2703.1

Mo X, Liu S, Lin Z, Guo R (2009) Regional crop yield, water consumption and water use efficiency and their responses to climate change in the North China Plain. Agric Ecosyst Environ 134:67–78

Mo X, Guo R, Liu S, Lin Z, Hu S (2013) Impacts of climate change on crop evapotranspiration with ensemble GCM projections in the North China Plain. Clim Chang 120:299–312. doi:10.1007/s10584-013-0823-3

Moiwo JP, Yang Y, Li H, Han S, Yang Y (2010) Impact of water resource exploitation on the hydrology and water storage in Baiyangdian Lake. Hydrol Process 24:3026–3039. doi:10.1002/hyp.7716

Niu G, Yang Z, Dickinson RE, Gulden LE (2005) A simple TOPMODEL-based runoff parameterization (SIMTOP) for use in global climate models. J Geophys Res 110:D21106. doi:10.1029/2005JD006111

Ozdogan M, Rodell M, Beaudoing HK, Toll DL (2010) Simulating the effects of irrigation over the United States in a land surface model based on satellite-derived agricultural data. J Hydrometeor 11:171–184. doi:10.1175/2009JHM1116.1

Playán E, Pérez-Coveta O, Martínez-Cob A, Herrero J, García-Navarro P, Latorre B, Brufau P, Garcés J (2008) Overland water and salt flows in a set of rice paddies. Agr Water Manage 95(6):645–658. doi:10.1016/j.agwat.2008.01.012

Qin DH, Chen Y, Li X (2005) Climate and environment changes in china (i): climate and environment changes in china and their projections. Science, Beijing, p 562

Rodell M, Velicogna I, Famiglietti JS (2009) Satellite-based estimates of groundwater depletion in India. Nature 460:999–1002. doi:10.1038/nature08238

Sacks WJ, Cook BI, Buenning N, Levis S, Helkowski JH (2009) Effects of global irrigation on the near-surface climate. Clim Dyn 33:159–175. doi:10.1007/s00382-008-0445-z

Scanlon BR, Fauntb CC, Longuevergne L, Reedy RC, Alley WM, McGuire VL, McMahon PB (2012) Groundwater depletion and sustainability of irrigation in the US High Plains and Central Valley. Proc Natl Acad Sci USA 109:9320–9325. doi:10.1073/pnas.1200311109

Siebert S, Döll P, Hoogeveen J, Faures JM, Frenken K, Feick S (2005) Development and validation of the global map of irrigation areas. Hydrol Earth Syst Sci 9:535–547. doi:10.5194/hess-9-535-2005

Sorooshian S, Li J, Hsu KL, Gao X (2012) Influence of irrigation schemes used in regional climate models on evapotranspiration estimation: results and comparative studies from California's Central Valley agricultural regions. J Geophys Res 117:D06107. doi:10.1029/2011JD016978

Sun H, Liu C, Zhang X, Shen Y, Zhang Y (2006) Effects of irrigation on water balance, yield and WUE of winter wheat in the North China Plain. Agric Water Mange 85:211–218. doi:10.1016/j.agwat.2006.04.008

Tang Q, Oki T, Kanae S, Hu H (2007) The influence of precipitation variability and partial irrigation within grid cells on a hydrological simulation. J Hydrometeorol 8:499–512. doi:10.1175/JHM589.1

Taylor RG, Scanlon B, Döll P, Rodell M, van Beek R, Wada Y, Longuevergne L, Leblanc M, Famiglietti JS, Edmunds M (2012) Ground water and climate change. Nat Clim Change. doi:10.1038/nclimate1744

Trenberth KE, Dai A, Rasmussen RM, Parsons DB (2003) The changing character of precipitation. Bull Am Meteorol Soc 84:1205–1217

Vörösmarty CJ, Green P, Salisbury J, Lammers RB (2000) Global water resources: vulnerability from climate change and population growth. Science 289(5477):284–288

Wang J, Huang J, Rozelle S, Huang Q, Blanke A (2007) Agriculture and groundwater development in northern China: trends, institutional responses, and policy options. Water Policy 9:61–74

Wang J, Mendelsohn R, Dinar A, Huang J, Rozelle S, Zhang L (2008a) The impact of climate change on china's agriculture. Agric Econ 40:323–337

Wang S, Shao J, Song X, Zhang Y, Huo Z, Zhou X (2008b) Application of MODFLOW and geographic information system to groundwater flow simulation in North China Plain, China. Environ Geol 55:1449–1462

Wang S, Song X, Wang Q, Xiao G, Liu C, Liu J (2009) Shallow groundwater dynamics in North China Plain. J Geogr Sci 19(2):175–188

Xiong W, Matthews R, Holman I, Lin E, Xu Y (2007) Modelling China's potential maize production at regional scale under climate change. Clim Change 85:433–451

Yang K, He J, Tang W, Qin J, Cheng C (2010) On downward shortwave and longwave radiations over high altitude regions: observation and modeling in the Tibetan Plateau. Agric For Meteorol 150:38–46

Zou J, Xie Z, Yu Y, Zhan C, Sun Q (2014) Climatic responses to anthropogenic groundwater exploitation: a case study of the Haihe River Basin, Northern China. Clim Dyn 42:2125–2145. doi:10.1007/s00382-013-1995-2